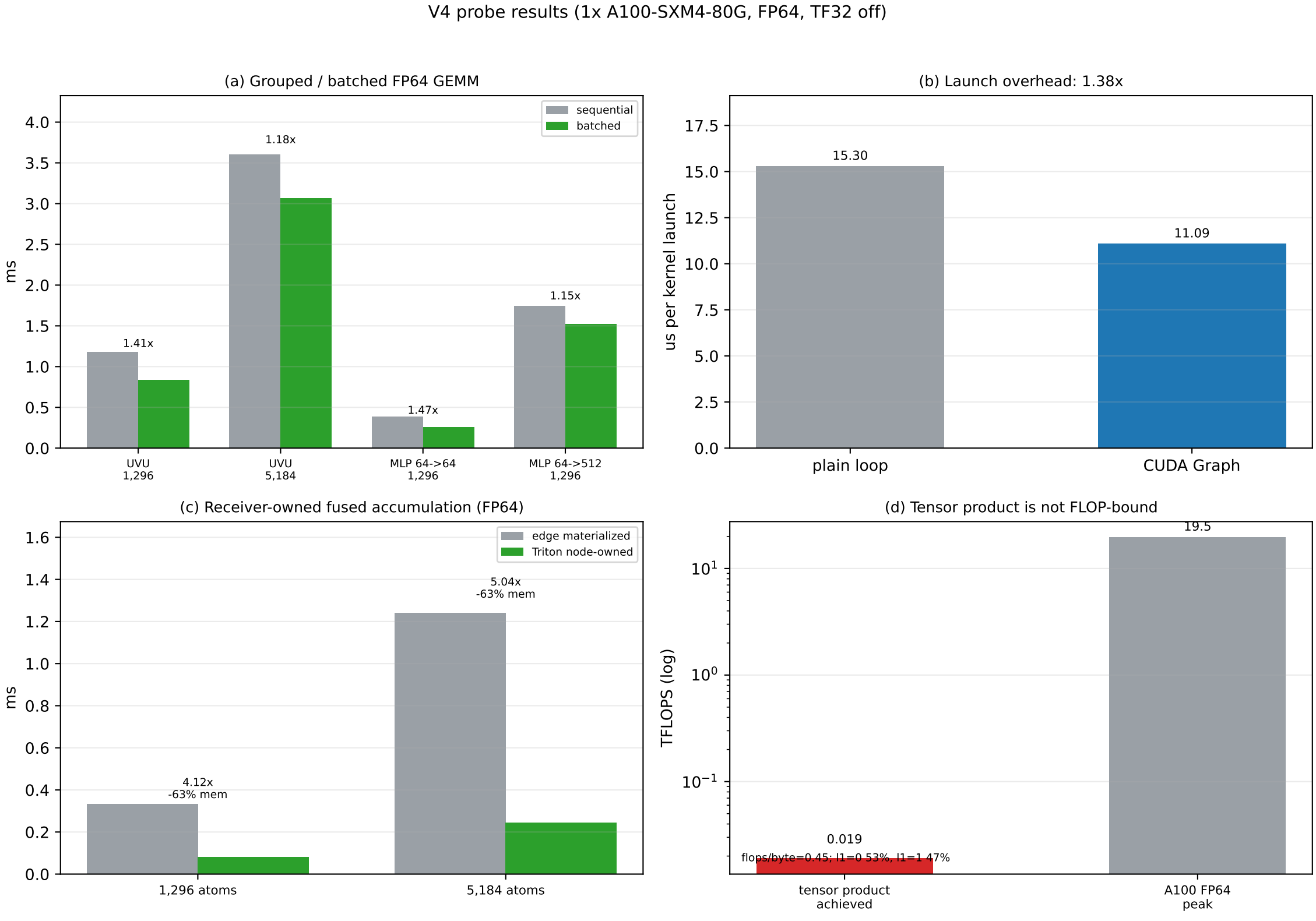
V4 probe results (1x A100-SXM4-80G, FP64, TF32 off)
(a) Grouped / batched FP64 GEMM
sequential
batched
ms
1.41x
1.18x
1.47x
1.15x
UVU
1,296
UVU
5,184
MLP 64->64
1,296
MLP 64->512
1,296
(b) Launch overhead: 1.38x
us per kernel launch
15.30
11.09
plain loop
CUDA Graph
(c) Receiver-owned fused accumulation (FP64)
edge materialized
Triton node-owned
ms
4.12x
-63% mem
5.04x
-63% mem
1,296 atoms
5,184 atoms
(d) Tensor product is not FLOP-bound
TFLOPS (log)
19.5
0.019
flops/byte=0.45; l1=0 53%, l1=1 47%
tensor product
achieved
A100 FP64
peak